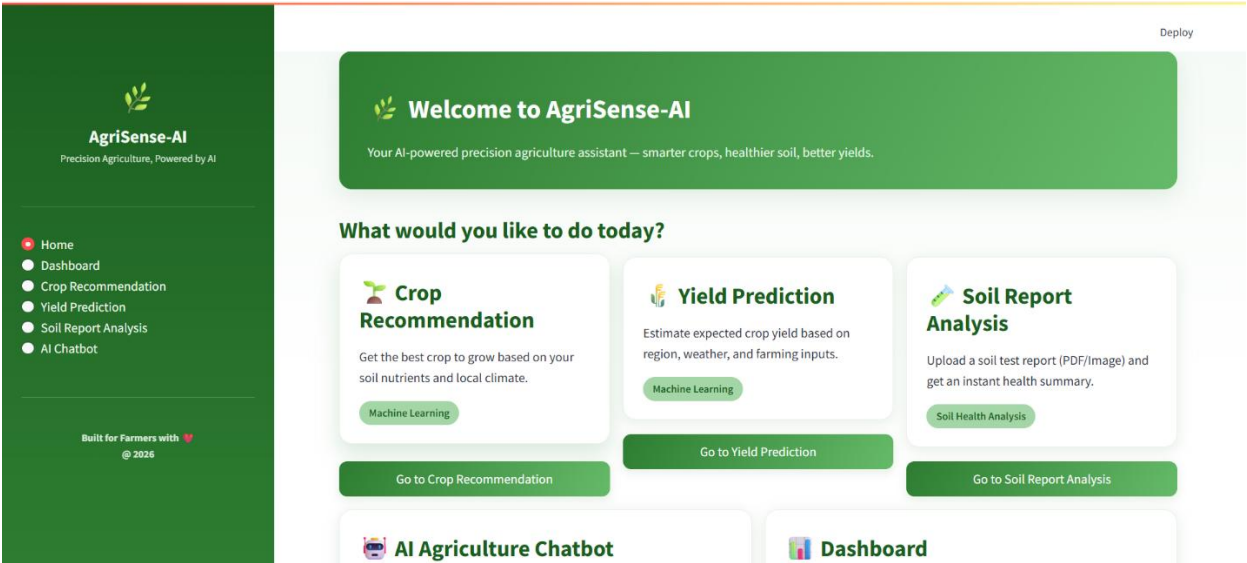


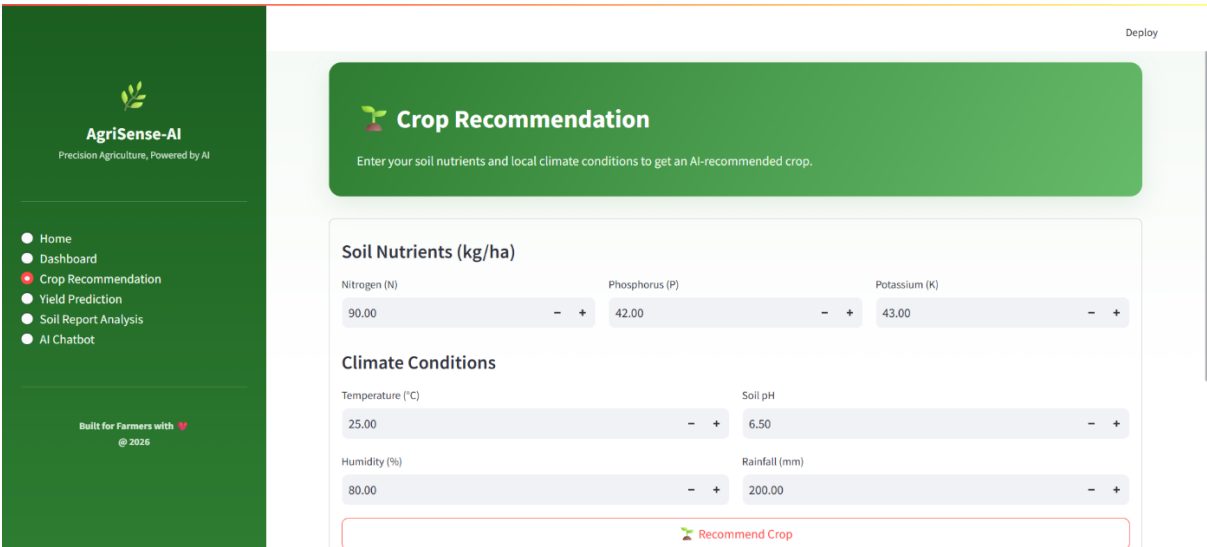
Working Application

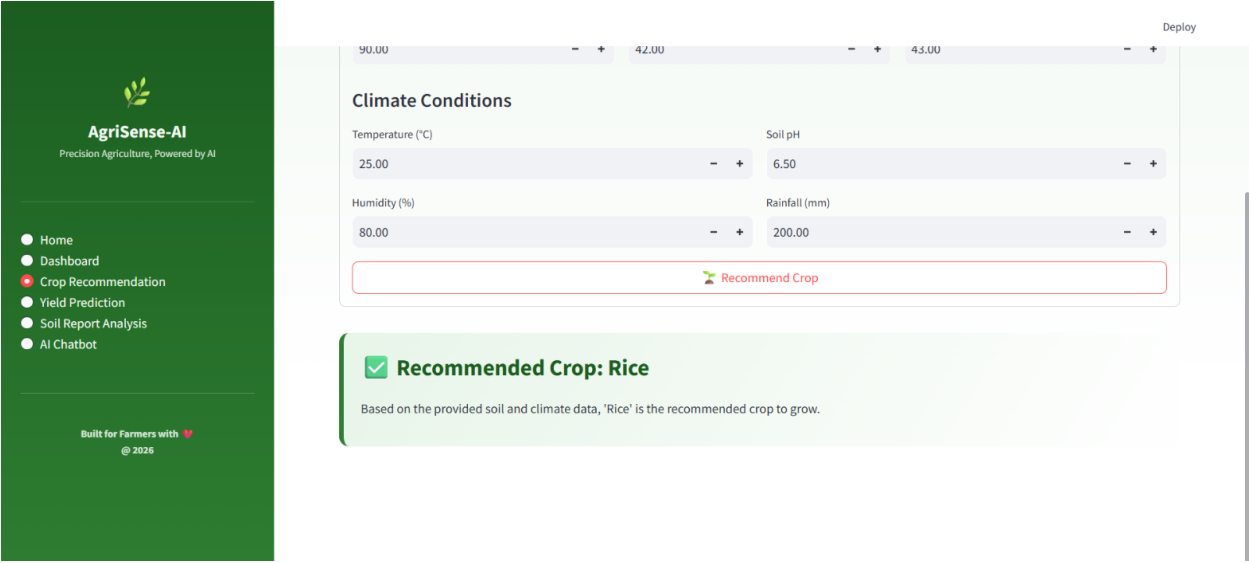
Home Page



Purpose: Serves as the landing page of the application and provides an overview of AgriSense-AI and its features.

Crop recommendation





Purpose: Uses the trained Random Forest Classifier to predict the most suitable crop and displays the recommended crop instantly.

Yield Prediction

**AgriSense-AI**
Precision Agriculture, Powered by AI

- Home
- Dashboard
- Crop Recommendation
- Yield Prediction**
- Soil Report Analysis
- AI Chatbot

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Deploy

**Crop Yield Prediction**
Estimate expected crop yield based on region, crop type, weather, and farming inputs.

Area / Region

India

Average Rainfall (mm/year)

800.00

Crop Item

Potatoes

Pesticide Usage (tonnes)

1000.00

Year

2026

Average Temperature (°C)

25.00

Predict Yield

 **Predicted Yield: 166501.06 hg/ha**

Purpose : Uses the Random Forest Regressor to estimate crop yield and Predicts expected agricultural production.

Soil Report Analysis

Enter Your Soil Test Values

Nitrogen - N (kg/ha)

245.00

Phosphorus - P (kg/ha)

18.00

Potassium - K (kg/ha)

130.00

pH

6.80

Organic Carbon (%)

0.45

Electrical Conductivity (dS/m)

0.60

Analyze Soil Health

 Analysis complete!

Nutrient Status

**245.0kg/ha**
NITROGEN
Low

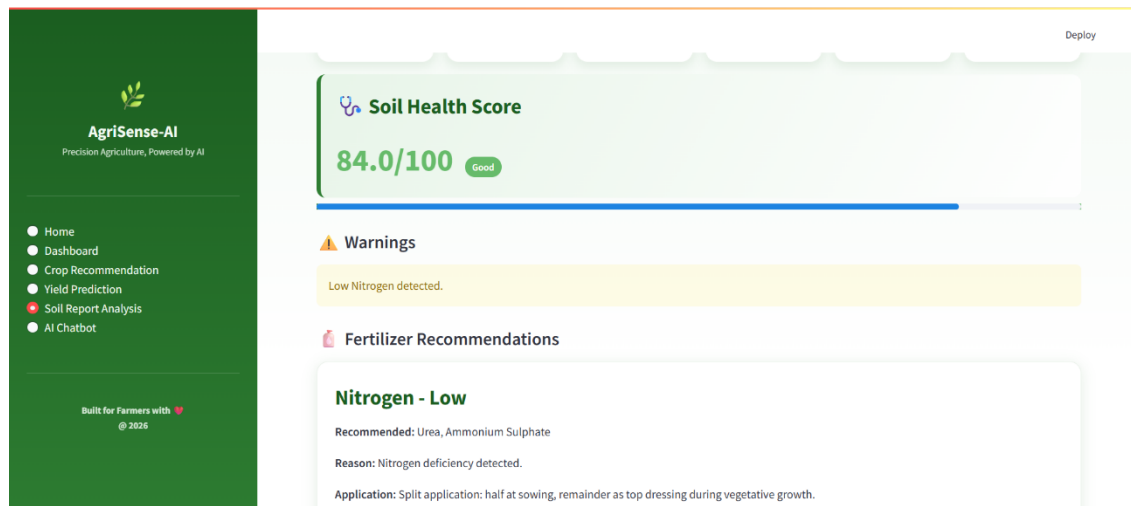
**18.0kg/ha**
PHOSPHORUS
Medium

**130.0kg/ha**
POTASSIUM
Medium

**6.8**
PH
Neutral

**0.45%**
ORGANIC CARBON
Optimal

**0.6dS/m**
EC
Optimal



Purpose: Allows users to manually enter soil test values and evaluates nutrient levels and overall soil health.

Agri-Bot (AI Chatbot)



AgriBot - AI Agriculture Assistant

Ask about soil health, fertilizers, crop selection, irrigation, and plant diseases.

💡 Example questions you can ask

👤 tomato cultivation



****Overview**** Tomato (*Solanum lycopersicum*) is a widely cultivated vegetable crop globally and in India, valued for its culinary versatility and demand in processing. It can be grown in open fields or protected structures like greenhouses. Ideal growth occurs between 20 and 27 degrees Celsius.

Explanation For tomato cultivation, seedlings are typically raised in a nursery for 25-30 days before transplanting. A seed rate of 300-400 grams per hectare is generally used for transplanted crops. Transplanting is best done in the evening or on a cloudy day to minimize transplant shock. Spacing varies by variety: determinate varieties require 60 x 45 cm, while indeterminate or staked varieties need wider spacing, around 75 x 60 cm.



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👤 How can I improve soil fertility?



****Overview**** Improving soil fertility is a key aspect of sustainable agriculture, and organic amendments play a crucial role in achieving this. Integrating such materials can enhance the soil's natural ability to provide nutrients for crops.

Explanation The application of Farmyard Manure (FYM) directly contributes to soil fertility by supplying essential nutrients. When FYM is incorporated into the soil, it lessens the dependency on chemical fertilizers by making nutrients available to plants. This is demonstrated by the ability to reduce chemical fertilizer nutrient recommendations for various crops when FYM is applied.

Best Practices To improve soil fertility, a recommended practice is the application of Farmyard Manure (FYM). The context specifies that applying FYM at a rate of 5 tonnes per hectare (5 tonnes/ha) can effectively reduce the need for specific chemical nutrients.

Example

- **For Rice:** If Farmyard Manure is applied at 5 tonnes/ha, the chemical fertilizer nutrient recommendation can be reduced by 25 kg Nitrogen (N), 5 kg Phosphorus (P_2O_5), and 12 kg Potassium (K_2O).
- **For Sugarcane:** Applying Farmyard Manure at 5 tonnes/ha can lead to a reduction of 25 kg N, 13 kg P_2O_5 , and 25 kg

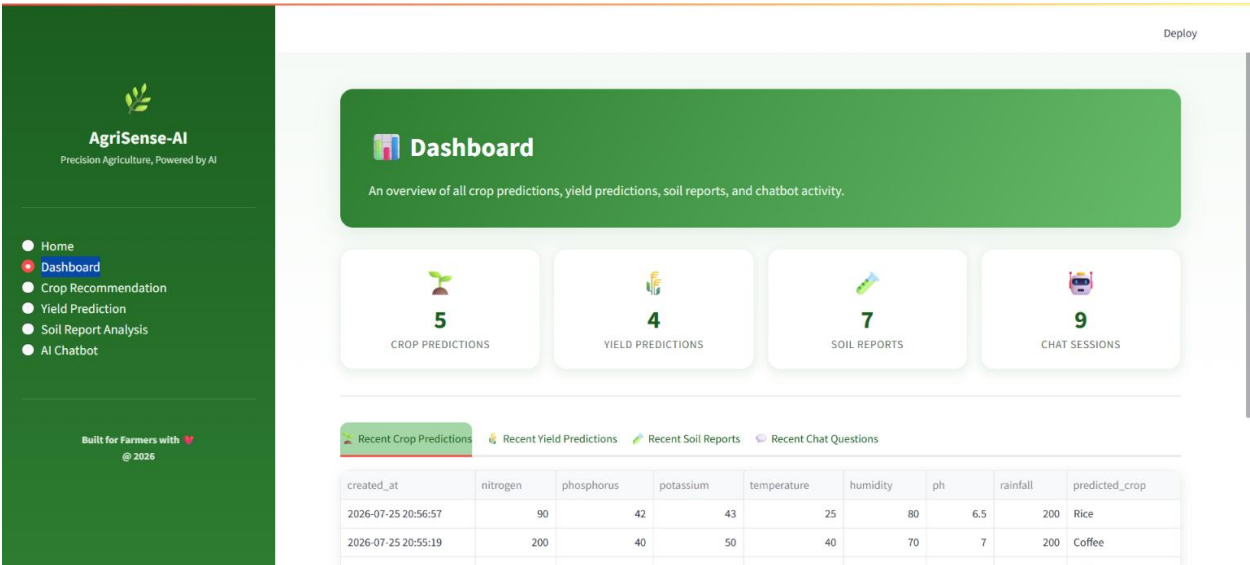
Ask AgriBot about soil, fertilizers, crops, irrigation...

Purpose: Accepts natural language questions related to agriculture.

- Retrieves relevant information from the agricultural knowledge base using LangChain + FAISS.

- Uses Google Gemini to generate context-aware responses.

Dashboard



Purpose : Acts as the central hub, allowing users to easily access and use all AI-powered agricultural services from a single interface.

